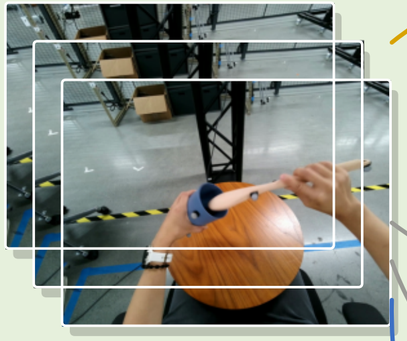
Egocentric Episode

raw video



colour camera + mono stereo pair

Task & Objects via VLM

Qwen3-VL (local)

"The person picks up a blue cup and a wooden spoon, stirs the contents of the cup with the spoon, and then places both items back on the table."

"wooden spoon", "blue cup"

Hand Estimator

Dyn-HaMR

metric MANO trajectory

WiLoR stereo fallback (<2%)

hands out of objects

$\mathcal{M}_t^h$



Metric Geometry

FoundationStereo



DROID-SLAM



table plane

gravity g

$D_t,\ T_t$

Object Estimator

SAM 3 (text-prompted)



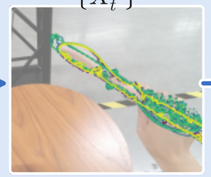
SAM 3D Objects + stereo scale

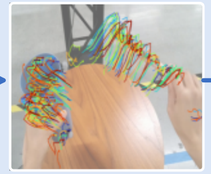


FoundationPose all ~252 hyp.

TrackEverything 3D tracks

$\{X_t^k\}$





$p_n(t)$

SelectTrack: interaction-aware refinement

Whole-sequence selection

$$\min_{k_{1:T}} \sum_t u_t(\mathcal{S}_t^o, X_t^{k_t}, p_n) + \lambda\, m_t(p_n)$$

Speed-limit flip filter

keep / flip / skip, ≤20° & 6 cm / frame

State-aware guards

$s_t \in \{\text{resting},\ \text{held}\}$

rest hold · spin guard · twist unwrap

Contact sanity

objects rest on the table
hands pushed out of objects

4D HOI Reconstruction



$\{\mathcal{M}_t^h,\ X_t^o,\ \mathcal{M}^o\}$

Robot Learning

Kaggle rollouts + meshes



Official evaluator

per-episode best run

FlashSAC per episode

task + contact-aware reward, 2 seeds

Retarget MANO → Sharpa



Simulator grounding

settled starts · carved collision · cup bases